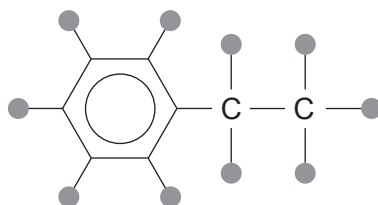


10. (a) Compound **B** contains only carbon, hydrogen and bromine. The hydrogen and bromine atoms could be at any of the positions indicated by a circle in the following structure.



- 5.35 g of compound **B** contains 3.74 g of bromine and 0.11 g of hydrogen
- It rotates the plane of plane polarised light
- When 4.36 g of compound **B** is heated with aqueous sodium hydroxide, the bromide ions produced react with silver ions to give 4.77 g of silver bromide ( $M_r$  187.9)

Use **all** of this information to deduce a possible structure for compound **B**.

Explain your reasoning.

[6 QER]

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- (b) Sucrose,  $C_{12}H_{22}O_{11}$ , is a sugar that contains eight —OH groups in each molecule. It reacts with ethanoic anhydride to give sucrose octaethanoate.



- (i) A 0.0700 mol sample of sucrose was used in a preparation.

Calculate the minimum volume of ethanoic anhydride needed to react with all the —OH groups present in this sample of sucrose. [2]

| ethanoic anhydride | $M_r$ | Density / $g\ cm^{-3}$ |
|--------------------|-------|------------------------|
|                    | 102   | 1.08                   |

Minimum volume = .....  $cm^3$

- (ii) There is interest in producing materials by 'green' methods.

Two methods are proposed for making sucrose octaethanoate in the laboratory.

Method X     Refluxing the mixture at 139°C using an electric hotplate

Method Y     Ultrasonic irradiation at room temperature

Apart from temperature, suggest **two** factors that should be considered when deciding which of these two methods is 'greener'. [2]

1. ....

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2. ....

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(c) Lactose is a reducing sugar and reacts with Fehling's reagent.

- (i) State what is seen when lactose reacts with the dark blue solution of Fehling's reagent. [1]

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- (ii) State which functional group must be present in a molecule of lactose for this result with Fehling's reagent. [1]

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